Deependra Singh

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.50
Saraswati Vidya Mandir I C Tirwa, Kannauj	Class XII (Board of High School and Intermediate Education, Uttar Pradesh)	2020	79.80%
Saraswati Vidya Mandir I C Tirwa, Kannauj	Class X (Board of High School and Intermediate Education, Uttar Pradesh)	2018	86.16%

PROJECTS

Hardware-Managed DRAM Cache for Tiered & Disaggregated Memory Systems | C++, ChampSim| Github

- Engineered a cycle-level DRAM cache controller managing a **128 MB HBM2 cache over 4 GB DDR4**, routing requests through a **10-state pipeline** with direct-mapped tag lookup and configurable near/far memory timing.
- Implemented **3 caching policies** over a **3-level queuing** (outstanding, conflict, write-back) and occupancy backpressure; benchmarked **42 configurations** across **6 SPEC CPU2006 workloads** (1.5B instructions each).
- Measured **~20% throughput loss** vs. no cache from **1.1-2.1× access amplification**; Oracle removed **20% traffic** but recovered only **4% IPC**, while **1 s CXL latency** caused a further **55% loss**, showing latency as the limiter.

Concurrent Transaction Processing Engine | C++, CMake, GoogleTest, Python | Github

- Built a multithreaded transactional database in C++ with **Strict 2PL**, wait-for-graph deadlock detection, and **OCC** via a unified concurrency-control interface.
- Validated correctness with **111 automated tests** and a **900-case concurrency stress matrix** across up to **32 threads**, with zero data races across **3 ThreadSanitizer runs**.
- Benchmarked both strategies across **150 configurations** (5 threads × 5 read/write ratios × 3 contention levels ×2 strategies) on identical workloads; **OCC achieved up to 7.3× higher throughput**.

Federated Weakly Supervised Video Anomaly Detection with Multimodal Prompt | Python, PyTorch, CLIP, Federated Learning | Github

- Developed a federated weakly-supervised video anomaly detector on a **CLIP vision-language backbone** with **multimodal prompts**, training **6 to 13 clients** across three federated data splits (Random, Scene, Event).
- Reproduced the **AAAI 2025** baseline, achieving **84.5 to 84.6 percent AUC** on UCF-Crime and **77.5 to 82.2 percent Average Precision** on XD-Violence across all splits.
- Designed **ABRATS**, a **Byzantine-robust** defense combining cosine-similarity update screening with EMA client-trust scoring to down-weight untrusted updates, layered with **trust-gated differential privacy** and gradient clipping.

Nyaya -- Multi-Agent Legal Reasoning System with GraphRAG & Symbolic Verification | Python, LangGraph, Neo4j/NetworkX, LLMs, FastAPI | Github

- Architected a **neuro-symbolic legal AI pipeline** combining dense embeddings, keyword-based retrieval, and **knowledge-graph traversal (GraphRAG)** across 810 statutory provisions spanning 17 Indian statutes.
- Integrated a **deterministic rule engine** to constrain LLM-generated legal arguments against codified statutory rules and flag **unsupported or fabricated legal citations**.
- Designed a **LangGraph-based multi-agent system** simulating adversarial legal proceedings with automated citation verification and trace-driven reasoning enabling an **auditable path from case facts to final decisions**.

SKILLS

Programming: C++, C, Python

AI and Machine Learning: PyTorch, Scikit-learn, NumPy, Pandas, OpenCV, CLIP, Federated Learning, RAG

CS Fundamentals: Data Structures & Algorithms, Operating Systems, Computer Architecture, Computer Networks, Database Management System

Systems/Tools: Linux, Git, Docker, ChampSim, FastAPI, Wireshark

Languages : Hindi, English

RELEVANT COURSES

- Foundations of Machine Learning - Quantum Computing
- Advanced Computer Architecture - Visual Computing
- Advanced Data Structures and Algorithms - Computer Networks

POSITION OF RESPONSIBILITY

Teaching Assistant -- Computer Architecture

- Assisted in conducting the course by supporting course activities, student queries, and academic coordination.

CERTIFICATIONS

Certificate of Computer System Security -- IIT Kanpur